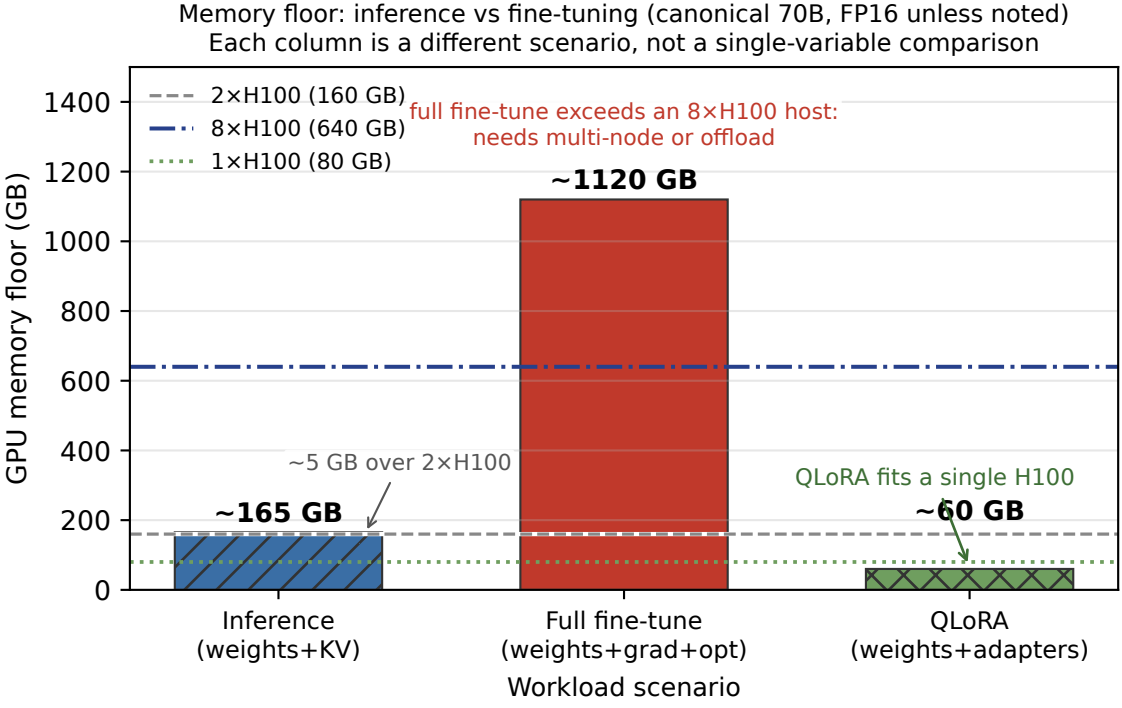

Each bar is a conditional floor (weights + the listed states), not a full deployment config.
Full fine-tune: FP16 weights + FP16 grads + Adam m/v/master (~16 B/param).
Activation, microbatch/sequence, and offload/checkpointing are NOT included (Table 7-2).
QLoRA: 4-bit (NF4) base + LoRA adapters + a small optimizer.
Specific adapter-rank value, activation memory, and offload/checkpointing are NOT included (Table 7-2).